

AI USE CASE INTEGRATION IN TEACHING-LEARNING

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-HC-03
2. **Use Case Title:** Urine Test Strip Reader using Colorimetry as a Point-of-Care Technology
3. **Application Domain:** Healthcare
4. **Collaborating Organization:** K. J. Somaiya Hospital and Research Center, Mumbai; Paramcare Lifesciences Pvt. Ltd., Mumbai
5. **Faculty Owner(s):**
 - Dr. Ninad Mehendale, Associate Professor, Department of Electronics Engineering, K J Somaiya School of Engineering
 - Dr. Jagannath Nirmal, Professor and Head, Department of Electronics Engineering, K J Somaiya School of Engineering
6. **Beneficiaries:** Patients undergoing routine urinalysis at point-of-care settings; clinical staff and pathologists at K. J. Somaiya Hospital; diagnostic product users of Paramcare Lifesciences Pvt. Ltd.; rural and resource-limited healthcare providers requiring objective, automated urine analysis without laboratory infrastructure.
7. **Problem Statement:** Urine test strips are widely used for rapid, low-cost screening of key health parameters (glucose, protein, pH, ketones, blood, leucocytes, and others). However, the conventional method of reading results relies on subjective visual comparison against a printed colour chart, leading to significant inter-observer variability and categorical (range-based) outputs instead of precise numerical concentrations. There is no existing affordable, portable system that can objectively capture, process, and quantify all 10 reagent pads on a urine strip simultaneously under controlled conditions and output exact continuous analyte concentrations in a hospital or point-of-care setting.
8. **Background and Context:** Urine dipstick analysis is one of the most frequently performed diagnostic tests globally, used to screen for conditions including diabetes, urinary tract infections, kidney disease, and liver disorders. Commercial strips (such as those manufactured by Paramcare Lifesciences Pvt. Ltd.) carry up to 10 reagent pads that undergo colour changes at 60–120 seconds after urine contact. The challenge in automated reading arises from the non-linear, multi-dimensional nature of the colour-to-concentration relationship: pad colours do not change uniformly or independently, and ambient light variation further corrupts readings. Smartphone-based solutions have been explored but suffer from uncontrolled illumination and lens distortion. This project proposes a controlled hardware enclosure (“black box”) with calibrated LED illumination and a Raspberry Pi camera to eliminate environmental variability, combined with advanced non-linear regression and Computer Vision algorithms to deliver precise, continuous analyte concentrations — a step beyond the categorical outputs of prior systems.
9. **Objectives:**
 - 1) To design and fabricate a controlled illumination enclosure with brightness-regulated LED strips and a Raspberry Pi camera module to eliminate ambient light variability during strip image capture.
 - 2) To develop a Computer Vision pipeline using OpenCV to automatically detect and spatially isolate all 10 reagent pads on a horizontally aligned urine test strip.
 - 3) To implement advanced non-linear regression models (multidimensional polynomial interpolation or deep learning-based regression) to map extracted RGB values of each pad directly to exact continuous analyte concentrations.

- 4) To automate the full workflow: strip placement → 120-second reaction wait → triggered image capture → pad RGB extraction → concentration computation → structured digital report.
- 5) To integrate results with patient ID and timestamp into a hospital-compatible database for time-series health monitoring at K J Somaiya Hospital and for product validation at Paramcare Lifesciences.

10. Solution Approach:

The system uses a Raspberry Pi 5 as the central edge computing unit. A custom-built opaque black box houses the strip holder, brightness-controlled LED strips providing uniform illumination, and the Raspberry Pi camera module. Upon strip insertion and a timed 120-second biochemical reaction, the system auto-triggers image capture. A Python-based Computer Vision pipeline (OpenCV) applies fixed-geometry bounding boxes to isolate each of the 10 horizontally aligned reagent pads and extracts average RGB values per pad. These multi-dimensional RGB vectors are passed to a trained non-linear regression model (multidimensional polynomial interpolation or a deep learning regression head) that maps directly to continuous numerical concentrations for each analyte (e.g., “Glucose: 100 mg/dL”, “Protein: 30 mg/dL”). Results are logged to a structured patient database. No smartphone camera or external laptop is required — the entire pipeline runs on-device, making the system a true standalone point-of-care diagnostic unit.



Fig. 1: Picture of current Prototype

11. Operational Challenges:

- 1) Strip brand variability: different manufacturers use different dye formulations, so calibration models must be re-trained per strip brand.
- 2) LED aging and calibration drift: LED brightness shifts over time, degrading colour accuracy; requires periodic recalibration.

- 3) Pad crosstalk and overlap: adjacent reagent pads may bleed colour into each other during image capture, requiring precise segmentation.
- 4) Non-linear, non-independent colour response: colour transitions across the 10 pads are interdependent (e.g., pH affects other readings), demanding multidimensional models rather than independent single-pad regression.
- 5) Clinical validation: correlation of device output with gold-standard laboratory urinalysis results at K J Somaiya Hospital to establish clinical accuracy thresholds.
- 6) Data security and patient privacy: ensuring encrypted transmission and storage of patient test results in compliance with the Digital Personal Data Protection (DPDP) Act 2023. Critical technical, operational, and implementation-related questions that must be resolved to design, deploy, and effectively utilize the proposed solution.

12. Dataset: <https://www.kaggle.com/datasets/ninadmehendale/paper-strip-reader>

13. High-Level Workflow:

A. Sample Preparation

- Patient provides urine sample; strip is dipped and placed in the device holder
- Biochemical reaction initiates across all 10 reagent pads

B. Timed Reaction Wait

- Raspberry Pi timer counts 120 seconds for full colour development
- System remains in standby; no image capture during reaction period

C. Controlled Image Capture

- Camera auto-triggers at 120 seconds inside the sealed black box
- Brightness-regulated LED strips provide uniform, calibrated illumination

D. Pad Segmentation (Computer Vision)

- OpenCV pipeline applies fixed-geometry bounding boxes to isolate each pad
- Average RGB values computed for each of the 10 pads

E. Concentration Prediction (AI & ML)

- Non-linear regression model maps 30-dimensional RGB vector to exact analyte concentrations
- Multidimensional polynomial interpolation or deep learning regression head

F. Report Generation & Data Logging

- Structured digital report generated (patient ID, timestamp, 10 analyte concentrations)
- Results encrypted and securely transmitted to hospital database for time-series tracking

G. Clinical Review

- Clinician or pathologist reviews flagged results
- Abnormal values highlighted against clinical reference intervals
- Device output used as decision support; confirmation by clinician required

14. Expected Outputs:

- 1) A working prototype of the Urine Test Strip Reader device (Raspberry Pi-based, self-contained in a black box enclosure).
- 2) A trained non-linear regression model delivering exact continuous concentrations for all 10 analyte parameters (glucose, protein, pH, ketones, blood, bilirubin, urobilinogen, nitrite, leucocytes, specific gravity).
- 3) A Python software stack for image capture, Computer Vision pad segmentation, and ML inference, deployable on Raspberry Pi 5.

- 4) A patient-linked digital report (PDF/structured data) with 10 analyte values, timestamp, and patient ID.
- 5) A hospital-compatible database module for time-series urinalysis logging at K. J. Somaiya Hospital.
- 6) A published research paper validating device accuracy against gold-standard laboratory urinalysis results.

15. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	High	Patient urine test results are sensitive medical data. Implement AES-256 encryption for data at rest and in transit. Apply patient data anonymisation for model training datasets. Enforce role-based access control for hospital database. Comply with the Digital Personal Data Protection (DPDP) Act 2023.
AI Security & Trustworthiness	High	The regression model must produce consistent, reproducible outputs across different strip batches and lighting conditions. Validate model predictions against gold-standard laboratory results. Establish uncertainty quantification so clinicians are aware of model confidence levels.
Responsible AI	High	The AI model must not produce false negatives for critical analytes (e.g., glucose, blood). Ensure the system reports concentration ranges clearly flagged against clinical reference intervals. Present results as decision support only. Include clear uncertainty/confidence indicators in reports.
Generative AI / LLMs	Low	Not directly applicable to the core colorimetric pipeline. Future scope: LLM-based clinical report narration or anomaly explanation may be explored as an add-on feature.
Agentic AI	Low	Not applicable at present. The system follows a deterministic, fully automated pipeline with no autonomous agent required.
Human-in-the-Loop Decision Making	High	The device produces medical diagnostic data. A qualified clinician or pathologist must review and approve all results before any clinical action is taken. Flag any out-of-range results prominently for immediate clinical review.
Regulatory / Governance Considerations	High	The device qualifies as an In Vitro Diagnostic (IVD) medical device under Indian regulations (CDSCO MDR 2017). Compliance with CDSCO IVD classification and approval process is required before clinical deployment. Patient data handling must comply with the DPDP Act 2023. Ethics committee approval from K. J. Somaiya Hospital is required for clinical validation studies.
